Global Fuel Prices & Taxation (2020–2026)

Project Report: Data Analytics with Tableau

Prepared by: Divyansh Nayak | MediCaps University, Indore

1. Introduction

1.1 Project Overview

Fuel prices influence household expenses, transport costs, inflation and business planning. This project analyses a global dataset of petrol, diesel and LPG prices alongside Brent crude prices, tax percentages, region, income level and subsidy level from 2020 to 2026. Tableau visualisations make it easier to compare patterns across regions and countries.

- **Dashboard (Global Fuel Overview):** combines a global petrol-price map, regional comparisons, price trends, income-level comparison, taxation comparison and a taxation-versus-petrol-price scatter plot.
- **Worksheets:** regional petrol, diesel and LPG comparisons; fuel price trends from 2020–2026; fuel taxation comparison; petrol price by income level; and taxation versus petrol price.
- **Story:** a guided Tableau story that presents regional price differences, trends, countries with high taxation, and relationships between taxation and petrol prices.

Tableau Public dashboard link:

https://public.tableau.com/app/profile/divyansh.nayak/viz/Global_Fuel_Prices_Dashboard/GlobalFuelOverview?publish=yes

Tableau Public story link:

https://public.tableau.com/app/profile/divyansh.nayak/viz/Global_Fuel_Prices_Story/GlobalFuelPricesStory?publish=yes

1.2 Purpose

The purpose is to convert fuel-market data into understandable comparisons that help users examine how prices and taxation vary across regions and income groups, observe changes over time, and identify countries with comparatively high tax percentages. The visualisations support exploratory analysis rather than proving that taxation alone causes higher fuel prices.

2. Ideation Phase

2.1 Problem Statement

Fuel-price data contains multiple measures and country-level classifications. Without a consolidated view, it can be difficult to compare petrol, diesel and LPG prices, track changes over time, or understand how tax percentage and income level relate to price differences. The project focuses on three representative stakeholder groups:

Persona	Role	Need
Aarav Mehta	Transport / Fleet Planner	Compare regional diesel and petrol costs to support route and operating-cost planning.
Nisha Sharma	Business Analyst	Explore fuel-price trends, income-level differences and country taxation for reports.
Dr. Kavita Rao	Energy Policy Researcher	Review regional variation and taxation patterns to frame further policy questions.

2.2 Empathy Map Canvas

Quadrant	Fleet Planner	Business Analyst	Policy Researcher
Says	“Where are fuel costs highest?”	“What changed over time?”	“How do tax rates differ?”
Thinks	Fuel costs affect operating budgets.	Comparisons should be clear and repeatable.	Regional patterns need context.
Does	Compares regional price measures.	Filters charts and checks country values.	Reviews tax and price patterns.
Feels	Needs practical cost visibility.	Needs reliable, understandable findings.	Needs cautious evidence for discussion.

2.3 Brainstorming

Ideas considered and grouped by theme:

- **Geography:** map of petrol prices by country and comparisons across regions.
- **Fuel type:** compare average petrol, diesel and LPG prices by region.
- **Time:** track fuel-price movement from 2020 through 2026.
- **Taxation:** compare tax percentages by country and inspect their relationship with petrol prices.
- **Segmentation:** compare petrol prices across high-, middle- and low-income groups and use region/income filters.
- **Selected:** one overview dashboard and a guided Tableau story using the same set of worksheets.

3. Requirement Analysis

3.1 Customer Journey Map

Stage	User action	Dashboard support
Awareness	Opens the dashboard for a global overview.	Global petrol-price map and regional comparison charts.
Exploration	Filters or compares regions and income levels.	Region and Income Level filters on the dashboard.
Analysis	Checks trends, tax values and relationships.	Fuel Price Trends, Fuel Taxation Comparison and Taxation vs Petrol Price.
Decision	Identifies areas or countries for closer investigation.	Side-by-side comparisons and tooltips for selected marks.
Sharing	Presents the findings to others.	Published Tableau Public dashboard and story.

3.2 Solution Requirement

Functional requirements

- Display a global map of average petrol prices by country.
- Compare average petrol, diesel and LPG prices across regions.
- Show fuel-price trends from 2020 to 2026.
- Compare country-level tax percentages and petrol prices by income level.
- Provide interactive region and income-level filters and chart tooltips.

Non-functional requirements

- Readable chart titles, axes and labels with a consistent visual style.
- Responsive filtering and practical dashboard navigation.
- Public sharing through Tableau Public.

3.3 Data Flow Diagram

Global fuel dataset (Excel) → Tableau data source → field review and aggregation → worksheets and charts → Global Fuel Overview dashboard → Tableau story → Tableau Public sharing.

3.4 Technology Stack

Layer	Tool
Data source	Global fuel prices Excel dataset
Visualisation	Tableau Desktop
Publishing	Tableau Public
Documentation	Report and supporting project documents

4. Project Design

4.1 Problem Solution Fit

The project addresses fragmented fuel-price information by bringing country, region, income level, price measures and taxation into a single interactive analytical view. The dashboard supports comparison and exploration; interpretations should account for other influences such as crude-oil costs, subsidies, currency movements and local market conditions.

4.2 Proposed Solution

- **Global map:** geographical comparison of petrol prices by country.
- **Regional comparisons:** average petrol, diesel and LPG prices across seven regions.
- **Price trends:** monthly movement of petrol, diesel and LPG measures over 2020–2026.
- **Taxation comparison:** countries ranked by average tax percentage.
- **Income-level view:** compare petrol prices for high-, middle- and low-income groups.
- **Taxation vs petrol price:** scatter plot for exploring whether higher tax percentages align with higher petrol prices.
- **Story and delivery:** a guided sequence of findings published through Tableau Public.

4.3 Solution Architecture

Excel dataset → Tableau data source and field aggregation → individual worksheets (map, trend line, regional bars, taxation bars and scatter plot) → Global Fuel Overview dashboard → Global Fuel Prices Story → Tableau Public end users.

5. Project Planning & Scheduling

5.1 Project Planning

Phase	Activities	Output
1. Understanding	Define project purpose and business questions.	Business questions
2. Data preparation	Connect dataset and review fields, dates and measures.	Tableau data source
3. Visualisation	Build charts for regional, temporal and taxation analysis.	Worksheets
4. Dashboard	Arrange visualisations and add filters.	Global Fuel Overview
5. Story and publishing	Arrange story points and publish dashboard/story.	Public links
6. Documentation	Prepare report and supporting project materials.	Final report

6. Functional and Performance Testing

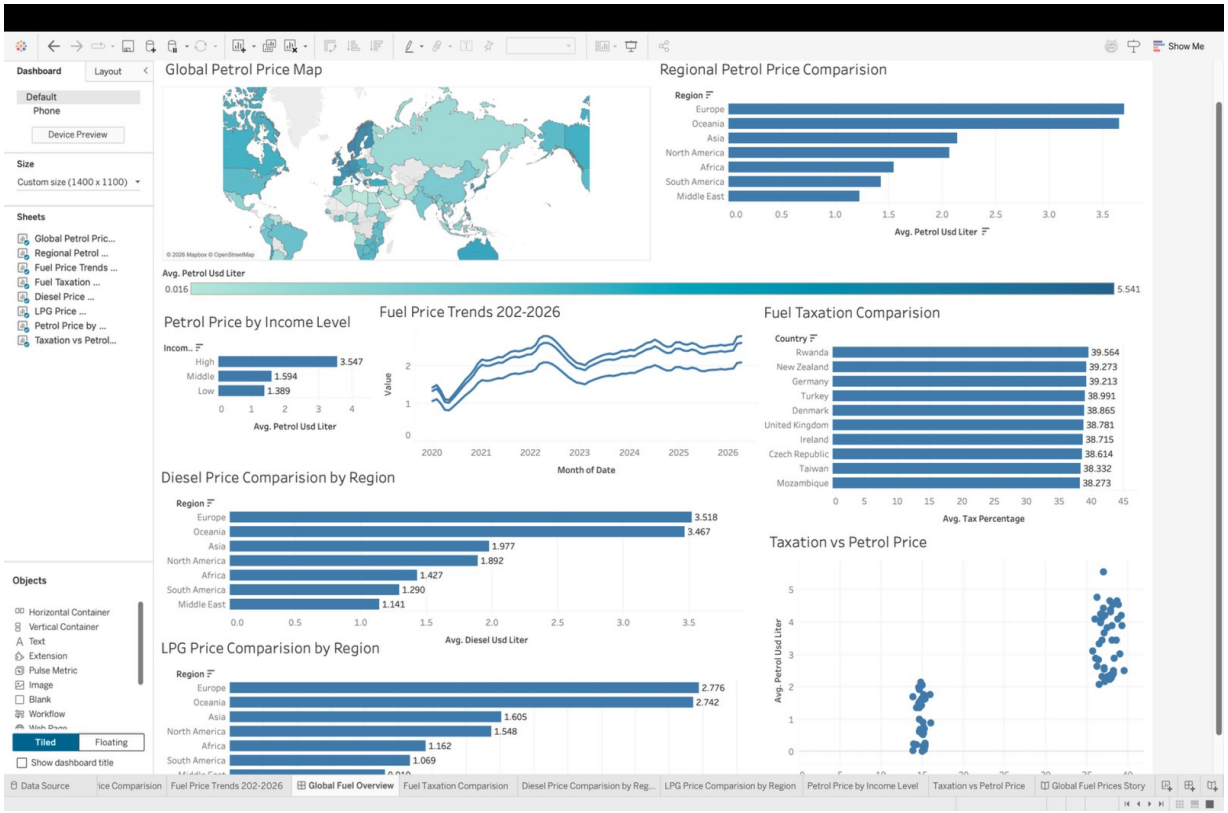
6.1 Performance Testing

Test	Expected result	Status
Regional price charts	Regions are compared using average fuel prices.	Reviewed
Taxation comparison	Country tax percentages appear in descending comparison.	Reviewed
Trend chart	Monthly trend across the available 2020–2026 period is displayed.	Reviewed
Dashboard filters	Region and income-level selections filter applicable views.	Reviewed
Tableau Public links	Dashboard and story open as published views.	Published

7. Results

7.1 Output Screenshots

Dashboard overview screenshot:



Key observations from the current Tableau views

- Europe and Oceania show the highest average regional petrol and diesel prices in the displayed comparisons.
- The displayed LPG comparison also places Europe and Oceania above the other regions.
- The income-level view shows the high-income group with a higher average petrol price than the middle- and low-income groups.
- The taxation comparison includes high values around 38–40% for several countries shown in the chart.
- The scatter plot displays variation in petrol prices across tax-percentage values; it should be treated as exploratory and not as proof of causation.

8. Advantages & Disadvantages

Advantages

- Combines geographic, regional, temporal and taxation comparisons in one Tableau project.
- Allows users to compare three fuel types and segment results by region and income level.
- Supports visual exploration through charts, filters and tooltips.
- The published dashboard and story are easy to share.

Disadvantages

- Results depend on the quality, coverage and definitions in the source dataset.
- Country averages may conceal within-country variation.
- The displayed associations do not establish that tax rates alone cause price differences.
- Tableau Public views are publicly accessible.

9. Conclusion

The Global Fuel Prices & Taxation project uses Tableau to compare petrol, diesel and LPG prices across regions, show trends from 2020 to 2026, and explore country-level tax percentages and income-level differences. The dashboard and story make the data easier to inspect and communicate. Findings are descriptive and should be interpreted alongside other economic and policy factors.

10. Future Scope

- Add more recent or regularly refreshed fuel-price data.
- Include inflation, exchange rates, crude-oil benchmarks and subsidy details in deeper comparisons.
- Add country-level time-series drill-downs and region-specific dashboard views.
- Use statistical analysis to test relationships between taxation and fuel prices.
- Improve mobile layout and add downloadable summaries.

11. Appendix

Source Code: Not applicable; the visualisations were created in Tableau.

Dataset: Global fuel prices Excel dataset used for the project.

Tableau Public dashboard: Open Global Fuel Overview

Tableau Public story: Open Global Fuel Prices Story